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A Novel Protocol for Evaluating Acid Etching Length and Fracture Conductivity in High-Temperature Carbonate Reservoirs

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Abstract

Acid fracturing remains a critical stimulation method for enhancing well performance in carbonate reservoirs by creating etched fractures that provide conductive pathways for fluid flow. The etched fracture length plays a pivotal role in determining treatment success. However, conventional methods for evaluating acid fracture conductivity, which rely on 7-inch long and 2-inch wide core samples tested using an API conductivity apparatus after acid etching, often fail to capture the full extent of acid etching behavior due to the limited length of conventional test samples. This study introduces a new systematic protocol designed to assess alternative acid systems' ability to transport live acid and achieve extended etched lengths under high-temperature conditions.

The experimental work utilized 4-inch diameter Indiana limestone cores, each 20 inches in length and sawed in half longitudinally, with permeability below 2 mD. Different acid systems were injected at a temperature of 200 °F for prescribed durations, and fracture conductivity was measured across a range of closure stresses. Effluent samples were analyzed for calcium ion concentration using inductively coupled plasma (ICP) to quantify rock dissolution, while computed tomography (CT) scans captured fracture morphology before and after acid injection. A successful procedure was developed which ensured the core sample integrity under closure stress similar to reservoir conditions. This new procedure is not only a more effective method to evaluate acid etching length and conductivity characteristics; it is also an easier and faster procedure compared with the traditional method.

The conductivity measured with the new procedure was compared with the results from the traditional procedure with similar conditions. The results of this work demonstrate that the novel protocol provides a robust framework for evaluating acid systems in fracture conductivity studies, addressing the limitations of traditional short-core experiments. These findings contribute to optimized acid fracturing designs tailored for challenging high-temperature reservoirs, enhancing treatment efficiency and reservoir productivity. The experimental observations and measurements can be used to calibrate predictive acid fracture models.

Introduction

Acid fracturing is a stimulation technique primarily applied in carbonate reservoirs to enhance hydrocarbon production, where a hydraulic fracture is first created by injecting fluid at pressures above the formation's breakdown pressure. Once the fracture is formed, acid is injected to dissolve portions of the fracture walls, generating conductivity through heterogeneous etching (Ruffet et al. 1998). Figure 1 illustrates a conceptual acid fracturing operation. The high dissolution rate of carbonate minerals in acid makes this method particularly effective in such formations. The primary mechanism of conductivity creation is the differential acid-rock reaction along the fracture faces, which leads to etched channels that remain open after pumping stops and the fracture closes (Cash et al. 2016). The effectiveness of acid fracturing is strongly influenced by etched fracture length, which in turn depends on the injected acid maintaining reactivity over long distances while minimizing rapid dissolution near the inlet (Navarrete et al. 2000). This mechanism is especially important in tight carbonate formations, where conventional proppant fracturing often fails due to poor proppant placement and embedment (Abass et al. 2006). In addition to total dissolution, the etching pattern also plays a critical role, with a channeling pattern created by acid injection tending to retain higher conductivity under closure stress compared to randomly distributed roughened fracture faces (Pournik et al. 2009).

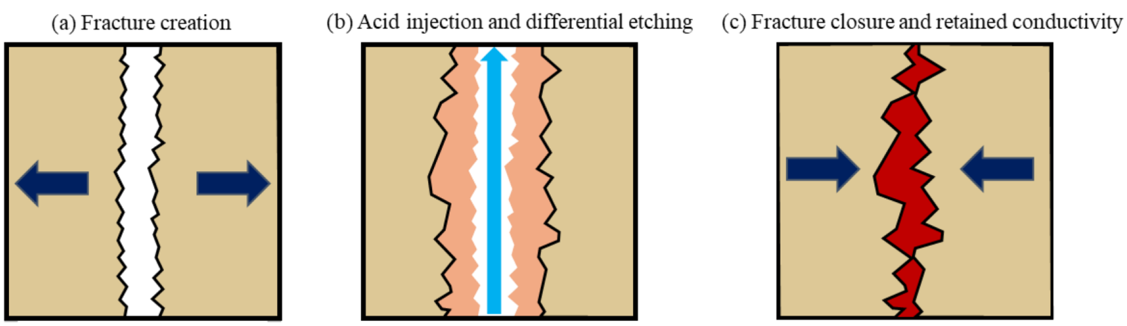

Figure 1—Conceptual illustration of an acid fracturing operation: (a) fracture creation by fluid injection, (b) acid injection and differential etching of fracture walls, and (c) fracture closure with residual conductivity due to etched channels.

Hydrochloric acid (HCl) is the most commonly used fluid in acid fracturing due to its strong reactivity with carbonate minerals and relatively low cost (Almubarak et al. 2022). However, in high-temperature formations, the rapid reactivity can be detrimental, as it causes excessive dissolution near the wellbore and limits live acid penetration length along the fracture surface. To overcome these limitations and promote deeper etching, different alternative acid systems have been developed with reaction retardation mechanisms. These include single-phase retarded acids (SPRA), which delay the acid-rock reaction through chemical buffering or diffusion control (Danican et al. 2024), and emulsified acids, which disperse acid in a hydrocarbon phase to limit mass transfer (Navarrete et al. 2000). These systems aim to improve acid transport, promote more uniform etching, and ultimately enhance fracture conductivity beyond what can be achieved with conventional HCl alone.

Laboratory evaluation of acid fracturing performance has long relied on short-core conductivity testing, with early studies using core samples approximately 1 inch in diameter and 2 to 3 inches in length (Nierode and Kruk 1973). These tests provide limited spatial resolution of acid-rock interactions. In the early 2000s, an improved procedure was developed which uses the API (American Petroleum Institute) standard 7-inch long, 2-inch wide fracture conductivity cell for testing. Acid etching is pumped through an open fracture between two core samples prior to placing the conductivity apparatus in a load frame for conductivity measurements.

Even with the improvement in testing sample length, the API procedure for acid-fracture conductivity still has a short distance between injection and outlet points, leading to a disproportionate emphasis on near-inlet dissolution. Short-core tests tend to overestimate the extent of etching in the early part of the fracture while providing little insight into the mid- and far-field performance of acid systems, limiting their usefulness in evaluating acid transport capacity and achieving extended etched lengths.

The acid resident duration in short-core experiments is often limited to seconds or a few minutes. This is particularly critical when evaluating alternative acid systems such as emulsified acids or chemically buffered retarded acids, which are specifically designed to maintain reactivity over extended distances. Short exposure times suppress the mechanisms that these acid systems are designed to leverage, leading to underestimation of their true transport capability and etching potential.

To improve the current practice of conductivity evaluation for acid fracturing, this study introduces a novel laboratory protocol that enables more representative evaluation of acid fracturing performance. The approach uses 20-inch-long, 4-inch-diameter carbonate core samples, sawed in half longitudinally. Polycarbonate shims, placed between the two core halves, are used to define a controlled fracture aperture and effective fracture height. Acid can be injected at elevated temperatures, and radial confining pressure is applied using a Hassler-type core holder to simulate fracture closure stress. The workflow incorporates pre- and post-treatment high-resolution CT imaging to analyze etched fracture morphology and effluent ICP analysis to quantify calcium dissolution. Fracture conductivity is then measured by injecting water through the etched core under varying confining pressures. Compared to traditional short-core API tests, this procedure is easier to implement and more representative for systematically evaluating acid transport behavior, etching effectiveness, and fracture conductivity under reservoir conditions. To ensure the reliability of this new method, the measured conductivity values were benchmarked against published data for Indiana Limestone, demonstrating consistency with established trends.

Development of the Novel Testing Protocol

To overcome the limitations of conventional short-core conductivity testing and better replicate field-relevant acid fracturing conditions, a new experimental protocol was developed.

Core Preparation and Fracture Configuration

In this study, the test specimens consisted of Indiana limestone outcrop core samples, chosen for their consistent mineralogy, low permeability (<2 mD), and widespread use in acid fracturing research. Each core measured 4 inches in diameter and 20 inches in length and was sawed in half longitudinally to create a single, planar fracture surface. Prior to acid injection, the core samples were saturated in 3 wt.% KCl brine under vacuum for 24 hours to ensure complete fluid saturation.

To define the fracture aperture and ensure structural stability, polycarbonate shims with a uniform thickness of 1/16 inch were placed between the sawn core halves. Two shim configurations were tested: one using two shims, and another using three. Depending on the configuration, the effective fracture height varied between 2 inches and 3.25 inches. These shims serve two purposes: (1) mechanical support during acid injection, which proved critical in early trials where cores fractured without adequate support, and (2) confining the acid flow through the intended fracture path, this is especially helpful when high-injection velocity is desired. To further prevent acid leakage along the sides, RTV silicone was applied to seal the outer edges of the core-shim assembly. This ensured that injected acid reacted primarily with the exposed fracture surfaces, improving both the consistency of etched patterns and the reliability of post-injection conductivity measurements. Figure 2 shows the two core/shim configurations for the tests. Overall, this configuration enabled stable acid injection and helped maintain core integrity during the high-pressure, high-temperature treatment.

Figure 2—Core samples prepared with the two shim configurations used in this study.

Test Apparatus and Acid Injection Procedure

The experimental study was conducted using a coreflooding system designed to evaluate acid-rock interaction under high temperature and pressure conditions. The apparatus consists of a pumping system, two accumulators for fluid storage, a core holder, a confining pressure system, a differential pressure transducer to measure pressure drop across the core, a back pressure regulator to control system pressure, temperature-controlled heating tapes wrapped around the core holder and inlet fluid lines, and a data acquisition system. A schematic of the full experimental setup is shown in Figure 3.

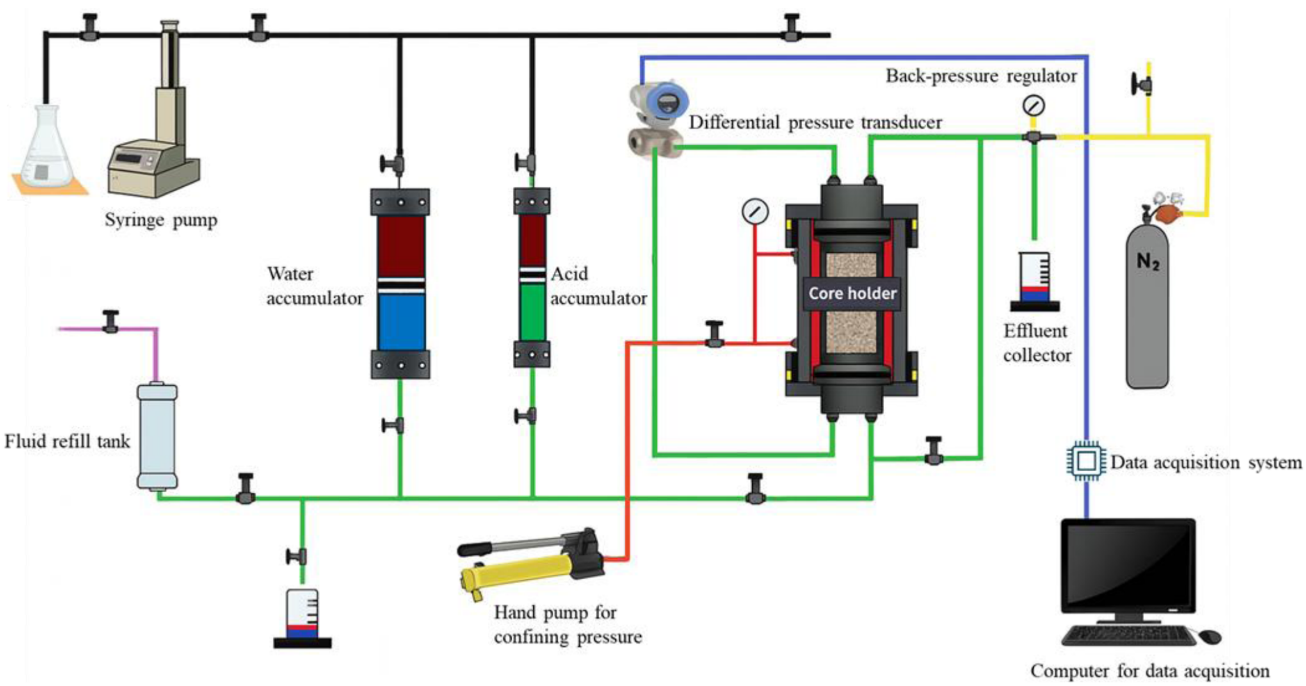

Figure 3—Schematic of coreflooding system used for acid injection and conductivity testing.

After the prepared cores were inserted in the core holder, each experiment began with pressurizing the system by injecting water through the core. A back-pressure regulator was used to maintain the desired minimum injection pressure. To maintain CO_2 in solution during acid injection and prevent degassing, a minimum back pressure of 1,000 psi was used.

In typical coreflood experiments with intact cores, the confining pressure is commonly set 400–500 psi above the injection pressure to prevent fluid bypass around the sample. However, this practice led to mechanical failure in the current study, as the cores were sawed longitudinally and thus structurally more vulnerable. We found that reducing the pressure differential and setting the confining pressure only slightly above the injection pressure (e.g., ~100 psi higher) was sufficient to prevent leakage while preserving core integrity throughout the acid injection period.

Once the system reached the target injection pressure and steady-state water flow was observed, the heating process was initiated. When both the fluid and the core holder reached the desired temperature, water injection was stopped and acid injection began. Acid was injected for a prescribed duration at a specified flow rate, and effluent samples were collected throughout the injection for subsequent ICP analysis to quantify rock dissolution. After the acid injection was complete, the core was flushed with water to displace any remaining acid and prepare the sample for removal and post-treatment evaluation.

Post-Acidizing Characterization

Following acid injection and water flushing, the core samples were removed from the core holder and subjected to high-resolution CT scanning. After imaging, the polycarbonate shims were detached. To eliminate the unreacted regions underneath the shims and minimize their impact on post-treatment analysis, the shim-contact surfaces were carefully trimmed using a combination of a sanding belt and a rotary Dremel tool.

High-resolution CT scans were performed before and after acid injection to capture changes in fracture geometry and quantify etched volume along the core. Rigid registration, a CT image alignment technique in which the datasets are spatially aligned using only rotation and translation, was applied to align the pre- and post-acid scans for accurate voxel-to-voxel comparison. Noise reduction and image filtering were used to enhance fracture contrast and minimize artifacts. A cylindrical mask was applied to isolate the core and remove empty surrounding space. The processed binary data were exported and further analyzed to calculate cross-sectional area changes along the etched fracture surfaces. Etched volume was calculated by segmenting the fracture region based on grayscale thresholding, measuring cross-sectional area across slices, and integrating over the core length. For each cross-sectional slice, the etched area was determined by calculating the difference in segmented rock pixels between the pre- and post-acid scans and multiplying that difference by the known in-plane pixel area. These area differences were tracked along the entire length of the core, providing a detailed etching profile. This approach enabled spatial mapping of acid-rock interaction and provided a quantitative basis for comparing the performance of different acid systems. Figure 4 shows representative CT images from an experiment.

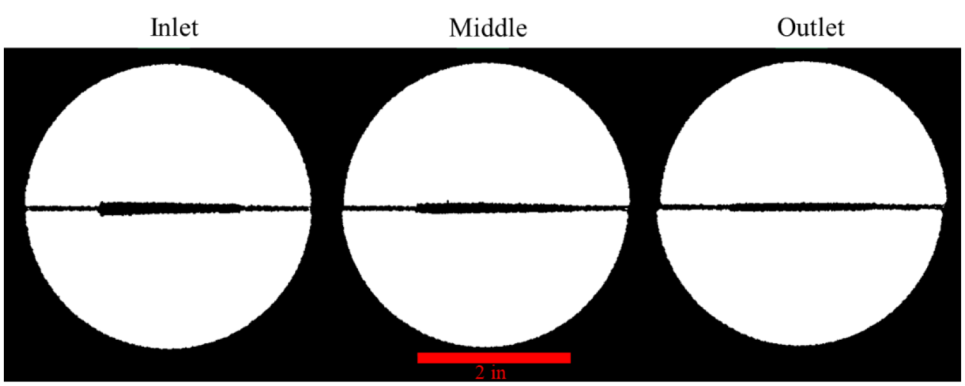

Figure 4—Representative post-acid CT scan slices at the inlet, middle, and outlet sections of the core, illustrating spatial variation in fracture etching.

Effluent samples collected during acid injection were analyzed to quantify calcium concentrations using ICP spectroscopy. Prior to analysis, each sample was diluted with deionized water to ensure measurements fell within the instrument's optimal detection range. ICP measurements provided the concentration of calcium ions (Ca^{2+}) in solution. The resulting calcium concentrations were used to estimate the mass of dissolved calcium carbonate ($CaCO_3$) based on a 1:1 molar stoichiometric relationship between Ca^{2+} and $CaCO_3$, assuming the primary dissolution product originated from the reaction:

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O \tag{1}$$

The mass of dissolved $CaCO_3$ was then calculated by converting the measured Ca^{2+} concentrations to moles and applying the molar mass of $CaCO_3$.

To evaluate the hydraulic conductivity of the etched fractures, the cores were wrapped in heat-shrink tubing to prevent fluid bypass, then reassembled in the core holder. Water was injected through the fracture at room temperature under steady-state flow conditions, and a back-pressure regulator maintained a constant pressure of 500 psi. Fracture conductivity was measured as a function of closure stress by incrementally increasing the radial confining pressure and recording the pressure differential across the core at each stage. At each confining pressure, water was injected at four different flow rates to validate Darcy flow behavior. After reaching steady-state conditions at each flow rate, the pressure drop was recorded and fracture conductivity was calculated using Darcy's law for linear flow in porous media, which is given by the equation:

$$q = \frac{k \quad A \quad \Delta P}{\mu \quad L} \tag{2}$$

where q is the volumetric flow rate [cc/min], k is the permeability, A is the cross-sectional area of flow [ft^2], ΔP is the pressure drop across the core [psi], μ is the fluid viscosity [cp], and L is the core length [ft]. In this study, flow is assumed to occur within the etched fracture created between the two longitudinally sawn core halves. The fracture cross-sectional area is defined as $A_f = h_f\, w_f$, where h_f is the fracture height and w_f is the fracture width.

Rearranging Darcy's law to solve for fracture conductivity (kw_f), the following expression is used:

$$kw_f = \frac{q \quad \mu \quad L}{\Delta P \quad h_f} \tag{3}$$

This formulation allows the direct calculation of conductivity from experimental data by measuring flow rate and pressure drop across the fracture at known fluid viscosity and fracture geometry (Arzuaga-García et al. 2024). The schematic shown in Figure 5 illustrates the flow path and geometry assumptions used in this calculation.

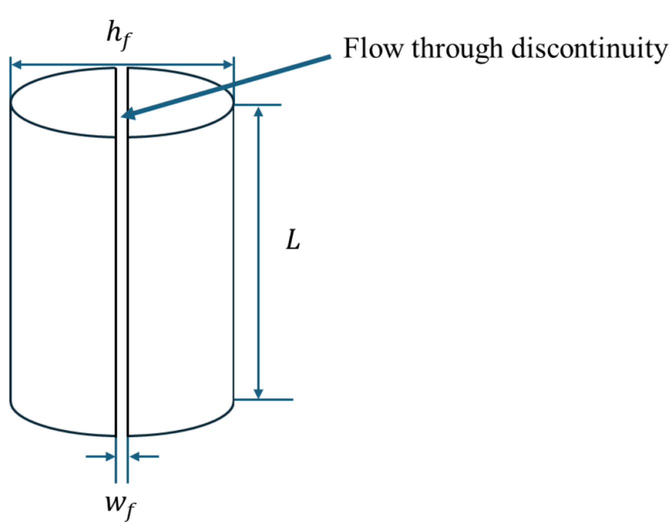

Figure 5—Schematic illustration of fracture geometry and flow path for conductivity measurements, showing fracture height, width, and length.

Results and Discussion

Two acid systems were evaluated in this study: (1) conventional hydrochloric acid (HCl) at 28 wt.% concentration with corrosion inhibitors and (2) an emulsified acid formulation prepared with a 70:30 acid-to-diesel ratio. For all experiments, acid was injected at 200 °F, and the total injected volume was fixed at 750 mL to enable direct comparison. However, the flow rate, and consequently the fluid velocity and residence time, was varied across the different experiments.

As described in the previous section, two shim configurations were used to control fracture geometry and provide structural stability during acid injection. These configurations created distinct flow paths and fracture heights, which directly impacted fluid velocity and residence time within the fracture. These differences in shim layout, flow area, and residence time are important when interpreting etching patterns and evaluating the performance acid systems. Moreover, these configurations enabled controlled investigation of how flow velocity and acid–rock contact time influence etching behavior under identical conditions.

In the three-shim configuration, polycarbonate shims (each 0.25 inches wide, 1/16 inch thick, and 20 inches long) were placed between the sawed core halves, two along the edges and one along the centerline. This configuration created a fracture aperture of 1/16 inch, an effective fracture height of 3.25 inches, and a total flow area of 1.31 cm^2. At a flow rate of 50 cm^3/min, the average fluid velocity within the fracture was approximately 38.15 cm/min with a residence time of 1.33 minutes, and a total contact area of 838.71 cm^2. Only the HCl acid system was tested in this configuration.

In the two-shim configuration, wider shims (each 1 inch wide, 1/16 inch thick, and 20 inches long) were placed along the outer edges of the core. This configuration created an effective fracture height of 2 inches and a smaller flow area of 0.806 cm^2. A higher flow rate of 75 cm^3/min yielded a velocity of 93 cm/min and a residence time of 0.55 minutes, with a total contact area of 516.13 cm^2. Both HCl and emulsified acid systems were evaluated in this configuration.

Throughout the Results and Discussion section, each test is referred to by its average fluid velocity (38 cm/min or 93 cm/min) for clarity, rather than by shim configuration.

Figure 6 shows post-acid core surface images from the three tests, HCl at 38 cm/min, HCl at 93 cm/min, and emulsified acid at 93 cm/min, highlighting variations in surface texture and etching morphology along the fracture face. In the high-velocity tests, HCl resulted in concentrated dissolution near the inlet, while the emulsified acid produced more heterogeneous etching marked by alternating bands of reacted and unreacted regions. Similar near-inlet etching was observed in the low-velocity test for HCl, where dissolution was again highly localized and diminished significantly along the flow path.

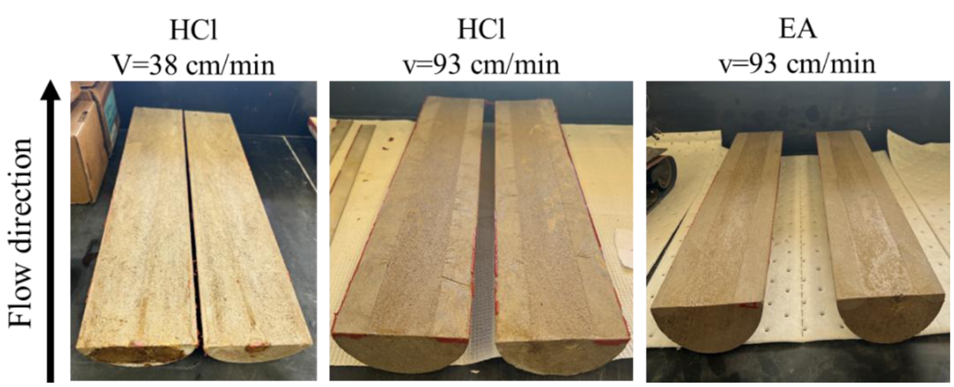

Figure 6—Post-acid core surface images, illustrating differences in surface texture and etching morphology.

Figure 7 presents effluent $CaCl_2$ concentration and cumulative etched volume as functions of cumulative injected volume. In both HCl tests, $CaCl_2$ concentrations (Figure 7a) peaked early, particularly at 38 cm/min, and declined rapidly, indicating dominant near-inlet dissolution. In contrast, the emulsified acid exhibited lower but more sustained $CaCl_2$ concentrations throughout the injection, indicating more gradual and distributed acid–rock interaction. Trends in etched volume (Figure 7b) were consistent with these observations, where HCl achieved higher early dissolution but plateaued sooner, while the emulsified acid maintained a steady increase. The total etched volume was greater for the 38 cm/min experiment because the fracture surface area was greater in that three-shim experiment.

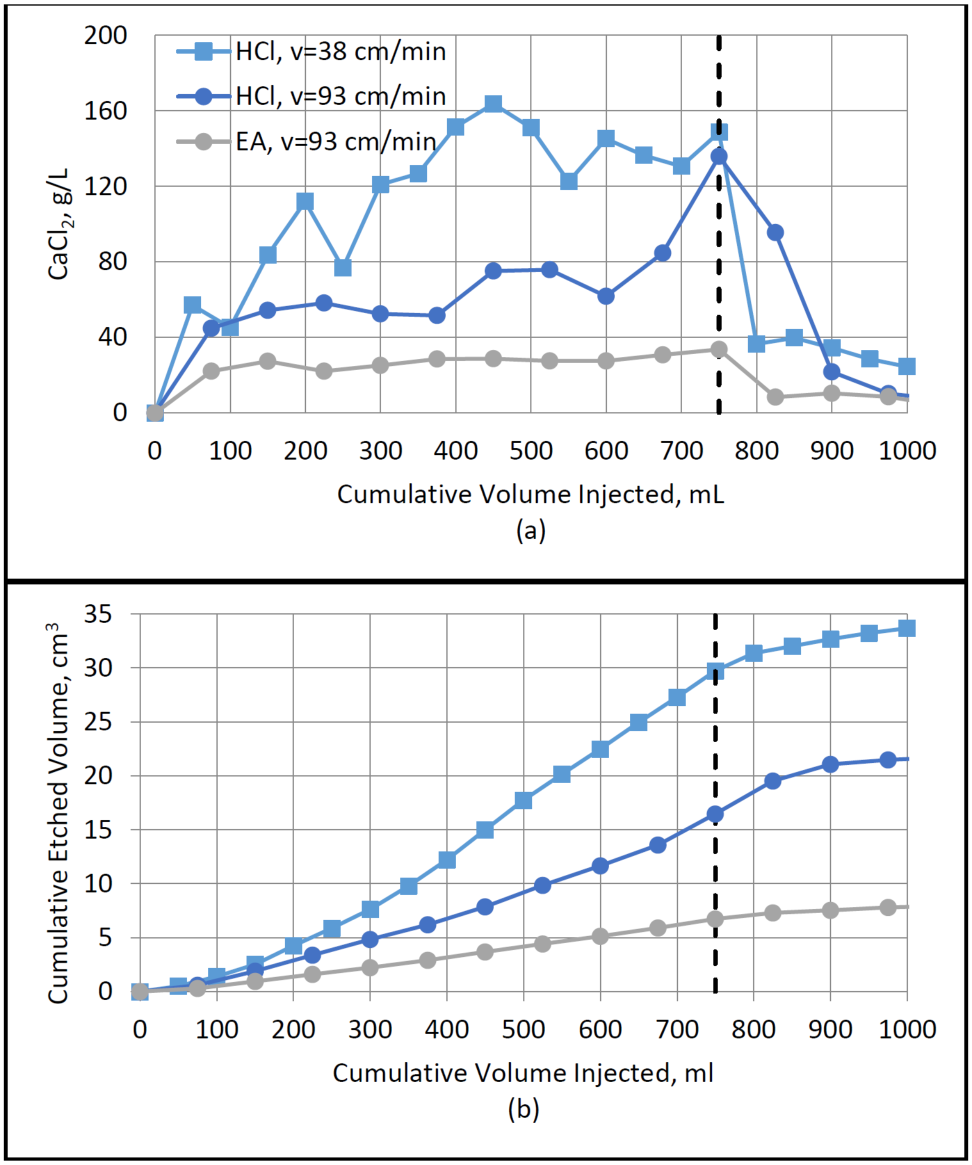

Figure 7—ICP analysis results: (a) effluent calcium chloride concentrations and (b) cumulative etched volume, both plotted as functions of cumulative volume injected. The dashed black line indicates the end of acid injection at 750 mL.

CT scan analysis further confirmed these differences. Figure 8 presents the etched volume profile as a function of distance from the injection face. HCl at 38 cm/min exhibited the highest initial etching near the inlet, followed by a sharp drop within the first 2–3 inches, after which the profile plateaued. At 93 cm/min, HCl showed slightly lower inlet etching but followed a similar downstream decay. In contrast, the emulsified acid produced a more uniform etching profile, with lower but consistent volume removal across the entire core length. These trends suggest that while HCl reacts more aggressively near the inlet, emulsified acid achieves deeper and more distributed penetration.

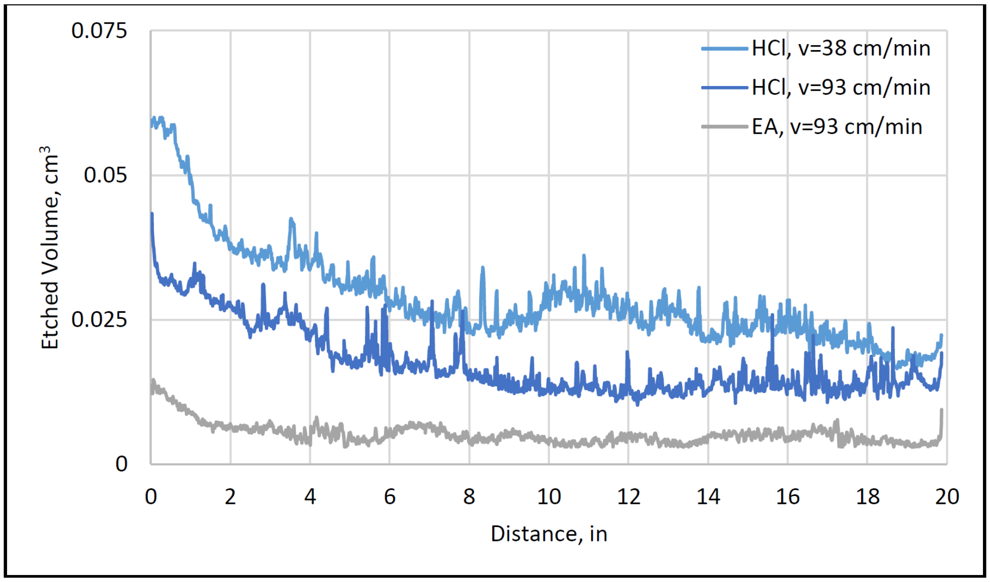

Figure 8—Etched volume profiles from CT scan, highlighting penetration depth differences across the three tests.

Fracture conductivity measurements are shown in Figure 9. As pressure increased, conductivity declined for all cases, with HCl at 38 cm/min showing the steepest reduction. The emulsified acid retained the highest conductivity at elevated pressures, particularly beyond 2,000 psi, demonstrating better mechanical integrity of the etched surfaces. These results suggest that while HCl may achieve initial high conductivity, its localized etching results in weaker structures that close more readily under stress.

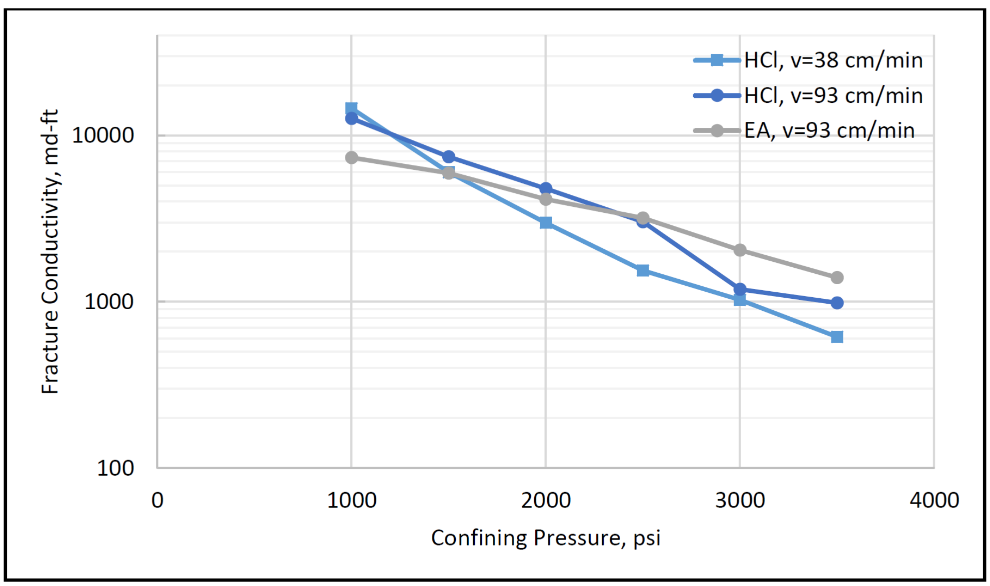

Figure 9—Fracture conductivity measurements as a function of confining pressure.

Figure 10 presents a comparison between ICP- and CT-derived etched volumes for the tests conducted, showing consistent trends and validating the complementary nature of the two techniques.

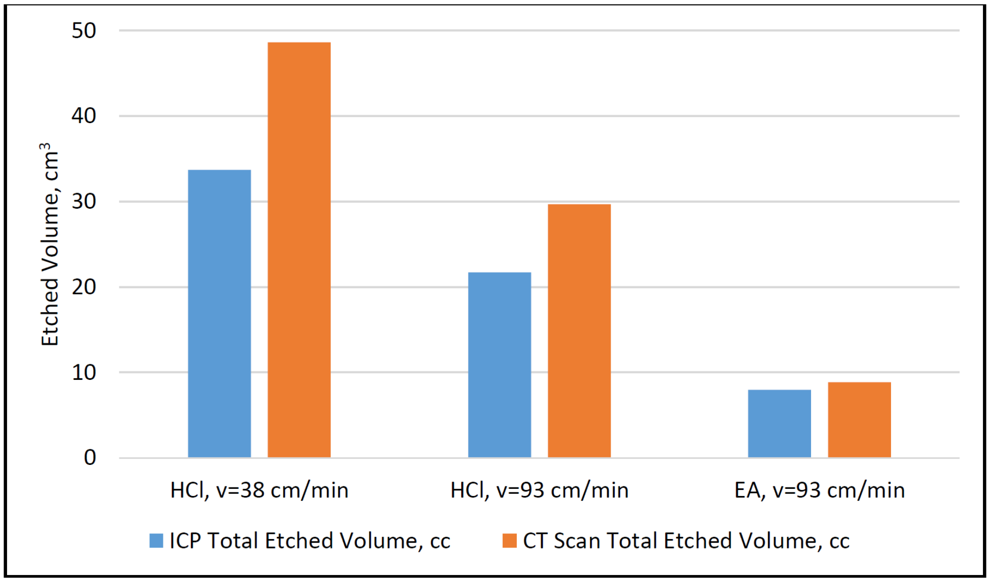

Figure 10—Comparison of total etched volume measured by CT scan and calculated from effluent (ICP) for each of the tests conducted.

A summary of the spatial etching profiles is provided in Figure 11, which compares the percentage of total etched volume occurring at various distances from the inlet for each test. The emulsified acid exhibited the lowest percentage of near-inlet etching (within 5 inches) and maintained a relatively high proportion of etching beyond 10 inches, indicating a more distributed and sustained acid-rock interaction. In contrast, HCl, particularly at 93 cm/min, showed more front-loaded behavior, with a greater share of etching occurring in the initial sections of the core. These results emphasize the value of transport control mechanisms for achieving deeper acid penetration, particularly under high-temperature conditions.

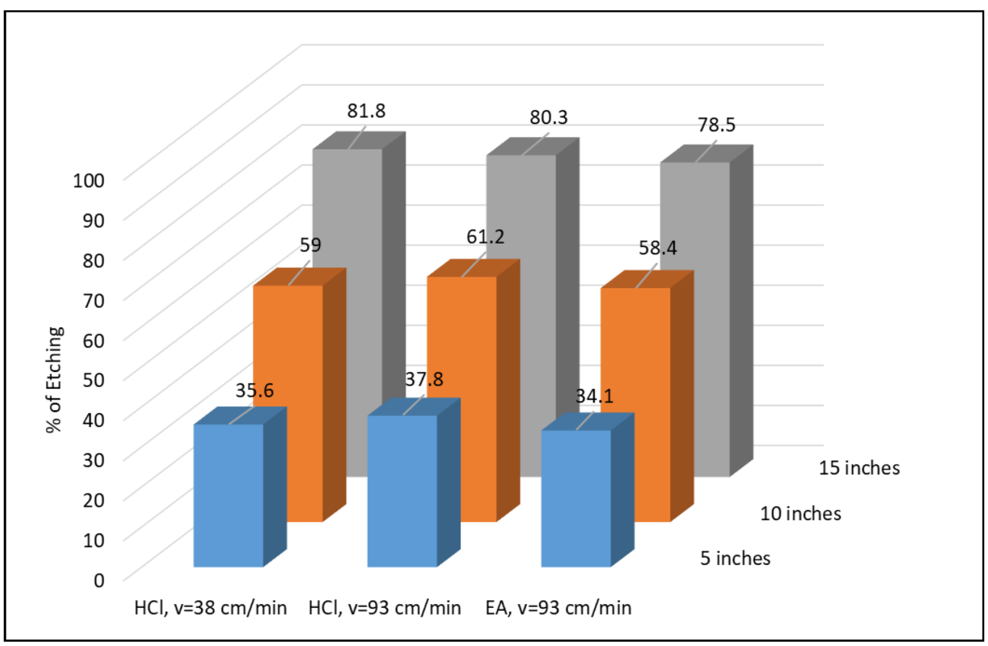

Figure 11—Summary of etching profiles as a percentage of total etched volume within 5, 10, and 15 inches of the fracture length for each of the tests conducted.

Protocol Validation and Benchmarking

The fracture conductivity values obtained in this study are consistent with previously published results for Indiana Limestone. For example, Pournik et al. (2007) reported fracture conductivity measurements on Indiana Limestone cores etched with different acid systems at the same temperature (200 °F) and under different contact times. In a subsequent study, Pournik et al. (2009) reported similar measurements with HCl at 15 wt.% but at a lower temperature (175 °F). Figure 12 presents the results from our experiments with selected data from their studies. The similarity in the range of conductivity values supports the validity of the experimental procedure and highlights the reproducibility of acid fracture conductivity testing on Indiana Limestone under comparable conditions.

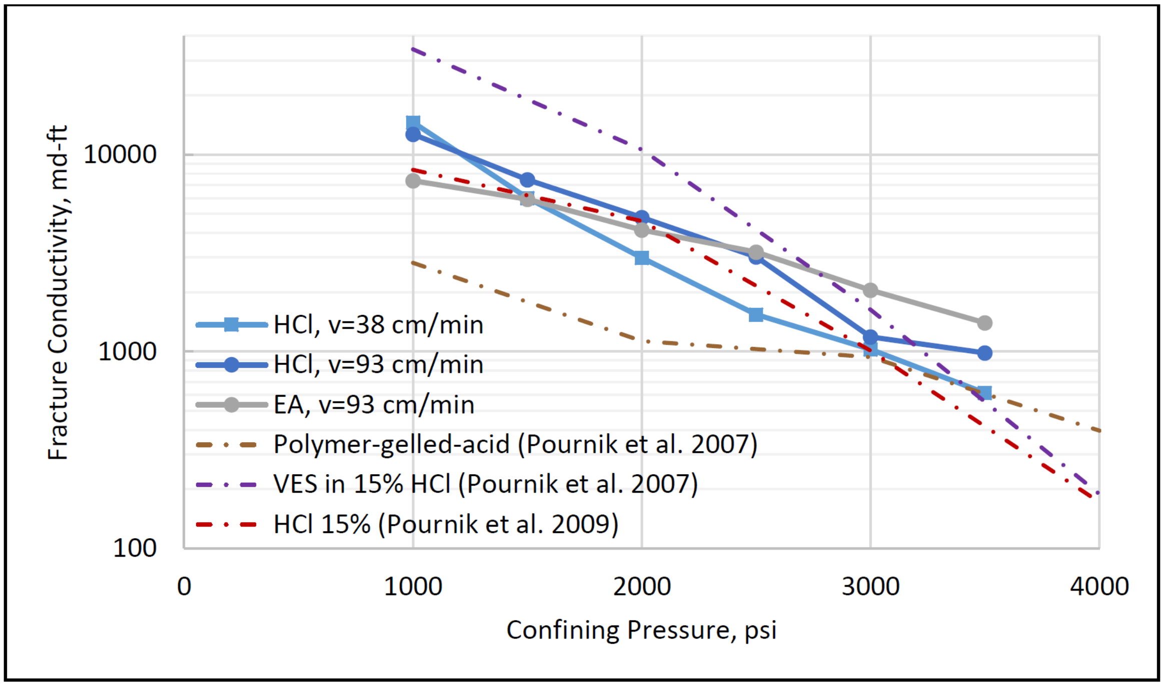

Figure 12—Comparison of fracture conductivity results from this study with those reported by Pournik et al. (2007, 2009) for Indiana Limestone etched with various acid systems.

Summary and Conclusions

This work introduces a novel protocol for evaluating acid etching length and fracture conductivity under high-temperature conditions. Using 20-inch-long carbonate cores and incorporating CT imaging, ICP analysis, and conductivity testing, the new protocol offered a more representative evaluation of acid system behavior compared to traditional short-core tests. The design enables assessment of spatial etching profiles, total rock dissolution, and conductivity retention after closure.

In the experiments conducted, the emulsified acid outperformed hydrochloric acid (HCl) in etching uniformity, calcium dissolution, and retained conductivity. HCl, tested at both 38 cm/min and 93 cm/min, exhibited localized inlet etching and sharp conductivity decline, whereas emulsified acid at 93 cm/min yielded more uniform etching and higher conductivity retention under stress. The results highlight the limitations of conventional HCl in long, high-temperature fractures and support the use of alternative acid systems for improved acid transport and fracture durability.

The method was further validated by comparing the measured fracture conductivities against published results for Indiana Limestone, showing consistency with established trends. While the protocol has limitations, including sleeve pressure constraints, core handling challenges, and resolution limits in CT and

ICP data, it provides a strong foundation for optimizing acid fracturing design. The protocol can also be extended to other rock types and acid systems, enabling broader applicability in acid stimulation design.

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